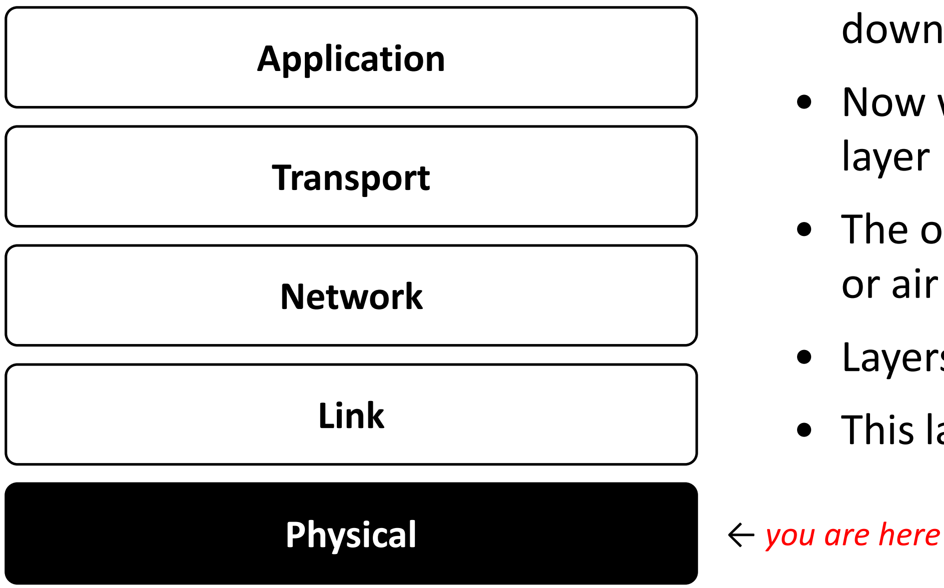
Chapter 2 · Week 2

The Physical Layer

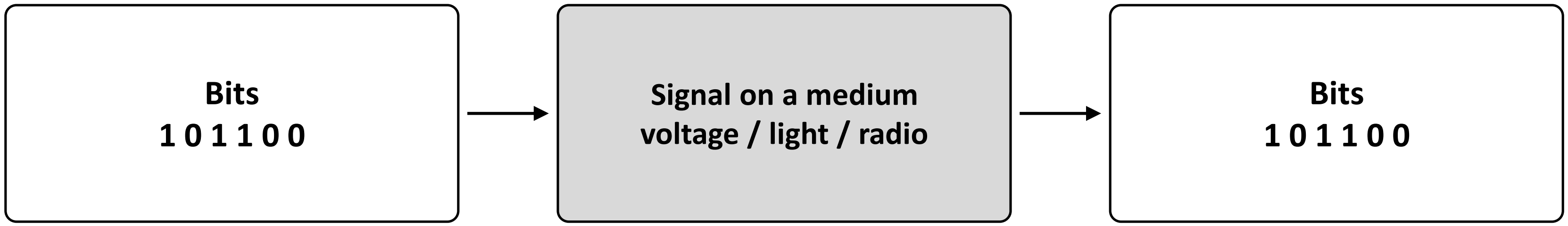
Transmission Media

Where we are in the stack



- Chapter 1 surveyed all five layers, top-down
- Now we go to the bottom: the Physical layer
- The only layer that touches copper, glass, or air
- Layers above assume a working bit-pipe
- This layer builds that pipe

What the physical layer does



- Carries a raw bit stream across a medium
- Turns bits into a physical signal, then back into bits
- Never interprets the bits — email, video, virus all look identical
- Only goal: deliver the bits and recover them cleanly

Two families of transmission media

Guided (wired)

- Signal confined to a conductor
  - Twisted pair
  - Coaxial cable
  - Optical fibre

Unguided (wireless)

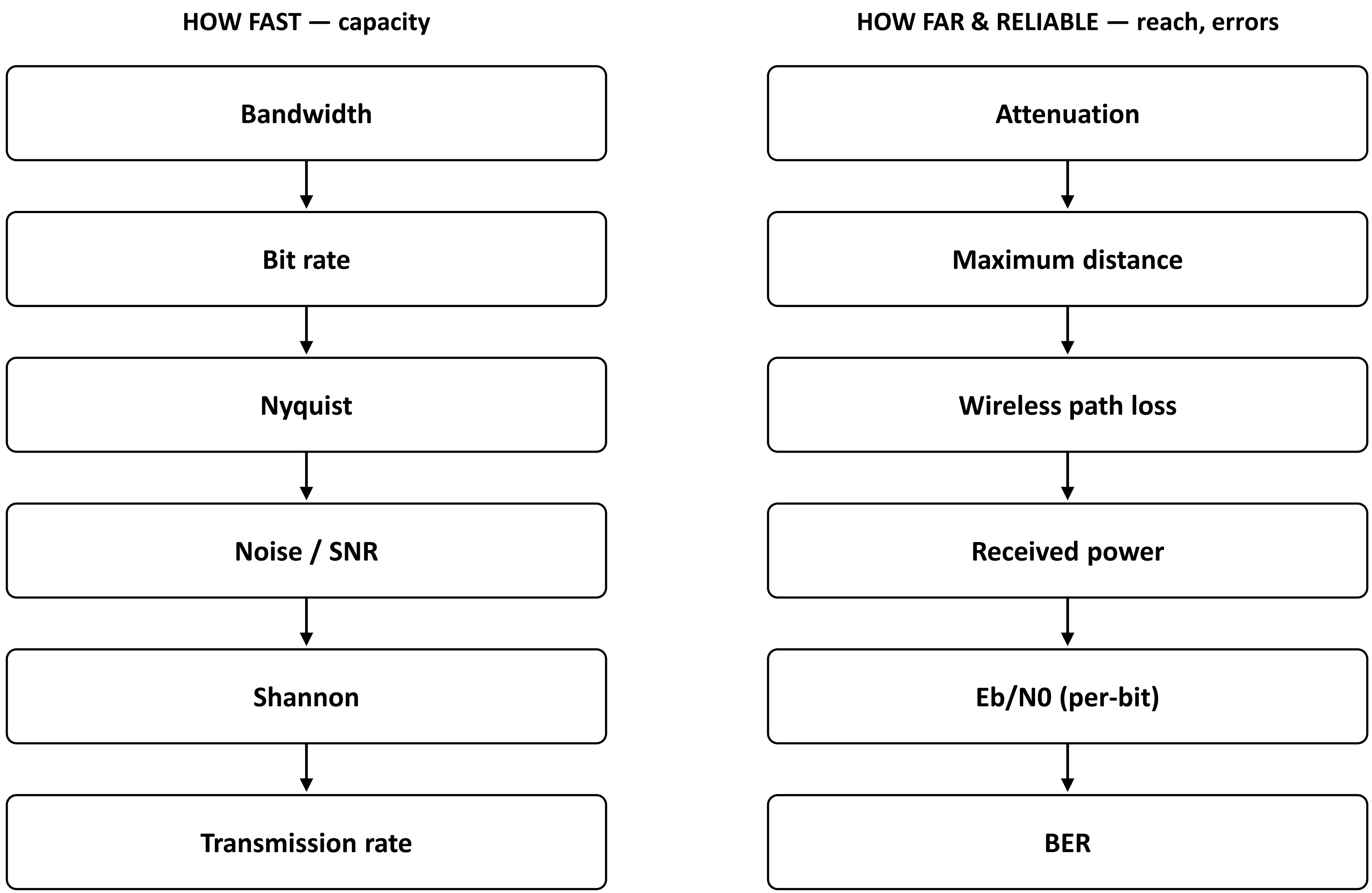
- Signal radiates through free space
  - Radio
  - Microwave
  - Infrared / light

How we judge a medium

- **Bandwidth — how much it can carry → data rate**
- **Attenuation — how far the signal survives → distance**
- **Noise / SNR — how clean the channel is → reliability**
- **Cost & deployment — cabling, right-of-way, maintenance**
- **Security / EMI — tapping risk, interference immunity**

The first three are physical quantities we can compute — the next slides show how.

The physical-layer toolkit



SNR (top) and Eb/N0 (bottom) are the same idea — channel-wide vs per-bit.

Bandwidth

- The range of frequencies a medium carries usably, in hertz
- The “width of the pipe”
- A wire is an accidental low-pass filter: passes lows, fades highs more and more steeply — not a wall
- The quoted number = where fading gets too bad to trust
  - Cat5 “100 MHz” ≈ 0–100 MHz; the top edge is a fuzzy cutoff, not a cliff
- **Width alone is not a bit rate — you must put bits in**
  - Bits are sent by varying the signal: at least 2 voltage levels (0 and 1); more levels carry more

Bit rate (and baud)

- **Bit rate = bits actually delivered per second (bps) — the number users see, e.g. “100 Mbps”**
- The medium carries a signal, not digits → encode bits as signal levels
- Symbol = one change of the signal to a new level
- Baud (symbol rate) = symbols per second
- Bits per symbol = $\log_2$(levels): 2→1, 4→2, 256→8
- **Bit rate = baud × bits per symbol**
  - Equal only at 2 levels; more levels → bit rate outruns baud

Nyquist- The noiseless ceiling

- Bandwidth B → at most 2B symbols per second (the baud limit)
- **Bit rate = 2B × $\log_2$(levels)**
- Cat5 (100 MHz) → 200M symbols/s:
  - 2 levels → 200 Mbps
  - 4 levels → 400 Mbps
  - 16 levels → 800 Mbps
  - 256 levels → 1.6 Gbps
- **Assumes a perfect, noiseless channel — an upper bound only**
  - Says nothing about whether the levels can be told apart

Noise and SNR

- Levels share a fixed voltage range → more levels sit closer together
- Noise randomly nudges the received voltage up or down
- Close-packed levels get smeared into their neighbours → wrong symbol
- **What matters: level spacing vs the size of the noise**
- **SNR (signal-to-noise ratio) = how far the signal stands above the noise**
  - High SNR → fine levels survive; low SNR → only coarse levels
- So noise caps how many levels are actually usable

Shannon — the real ceiling

- Prices in the noise that Nyquist ignored
- **$C = B \times \log_2(1 + SNR)$**
- Maximum error-free bit rate for a given bandwidth AND SNR
- **With noise present: Shannon < Nyquist, always**
  - No encoding scheme beats it
- Still idealized: assumes perfect, arbitrarily complex coding
  - Real hardware always lands a little under Shannon

Three ceilings, highest to lowest

Nyquist rate
the fantasy ceiling — no noise (e.g. 1.6 Gbps)

≥

Shannon capacity
the real physical cap — noise in, perfect coding assumed

≥

Achieved rate
actual hardware — always a little under Shannon

A datasheet's "Gigabit" is engineered safely below Shannon — not the Nyquist toy number.

Twisted pair

- The oldest, cheapest, most widely deployed guided medium
- Telephone lines; the Cat-5/6 Ethernet in this building
- **The “twist” fights interference: opposing twists cancel each other’s radiation**
  - More twists per cm → less radiation → more usable bandwidth
  - Cat3 → Cat5 → Cat6 is exactly that progression

Twisted Pair

- Waves from different twists cancel out , so the wire radiates less effectively. The more is the number of twists per cm lesser is the radiation.
- They run for several Km without amplification
- For longer distances repeaters are required.

Twisted Pair

(a)

(b)

(a) Category 3.

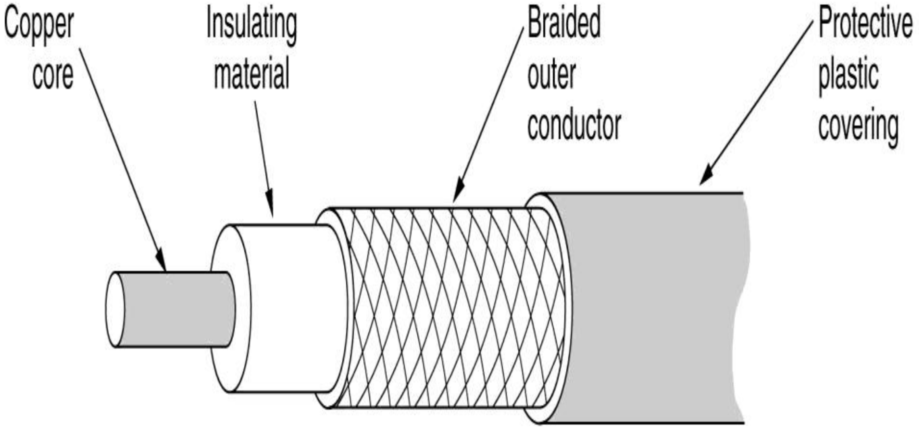
(b) Category 5.

Twisted Pair contd…

- Cat3 - 16 MHz
- Cat5 - 100 MHz
- Cat6 – 250 MHz for Gigabit Ethernets
- Cat7 – 600 MHz

- All these wirings are referred to as UTP (Unshielded Twisted Pair).

Coaxial Cables



Coaxial Cable contd..

- Better shielding hence better noise immunity
- High bandwidth upto 1GHz
- Earlier used on long distance telephone lines (short distance is twisted pair), now replaced with optical fibre
- Now used largely in cable TV and MANs.

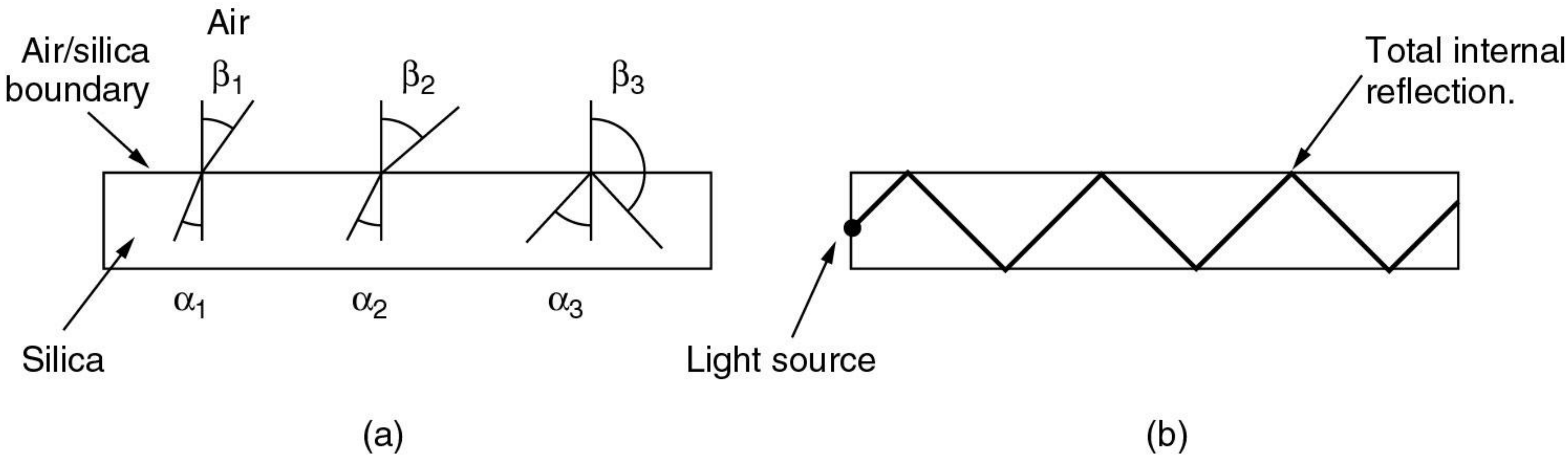
Fibre Optics

- 3 key components :
  - Light source : accepts an electrical signal,converts and transmits as light pulses.
    - LEDs
    - Semi-conductor lasers
  - Transmission medium
    - A very thin fiber of glass
  - Detector : senses the light pulses and converts it back to electrical signal

Fiber Cables (2)

Item	LED	Semiconductor laser
Data rate	Low	High
Fiber type	Multimode	Multimode or single mode
Distance	Short	Long
Lifetime	Long life	Short life
Temperature sensitivity	Minor	Substantial
Cost	Low cost	Expensive

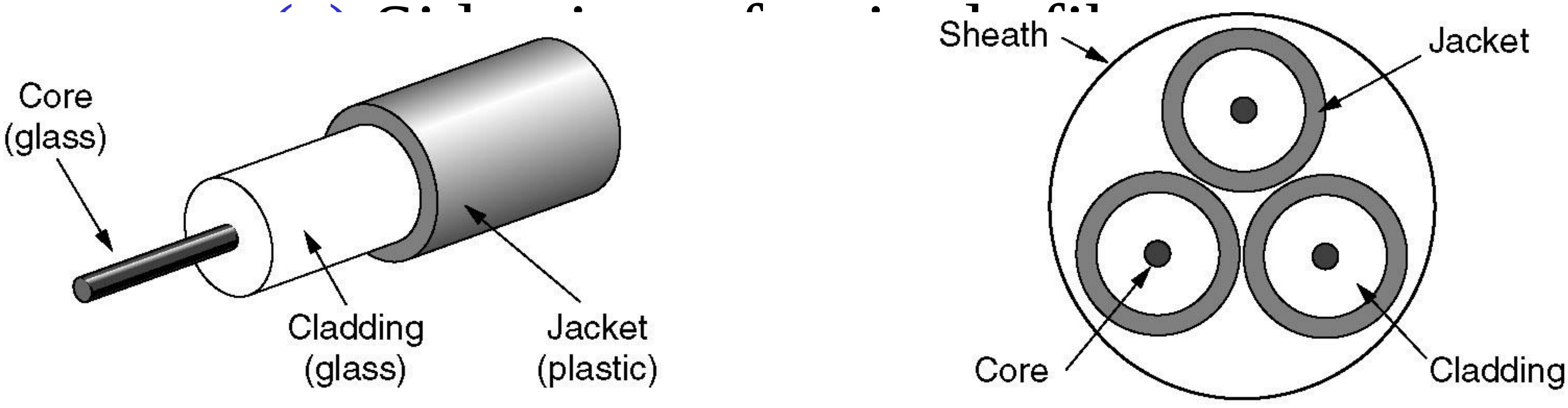
Fiber Optics



(a) Three examples of a light ray from inside a silica fiber impinging on the air/silica boundary at different angles.

(b) Light trapped by total internal reflection.

Fiber Cables

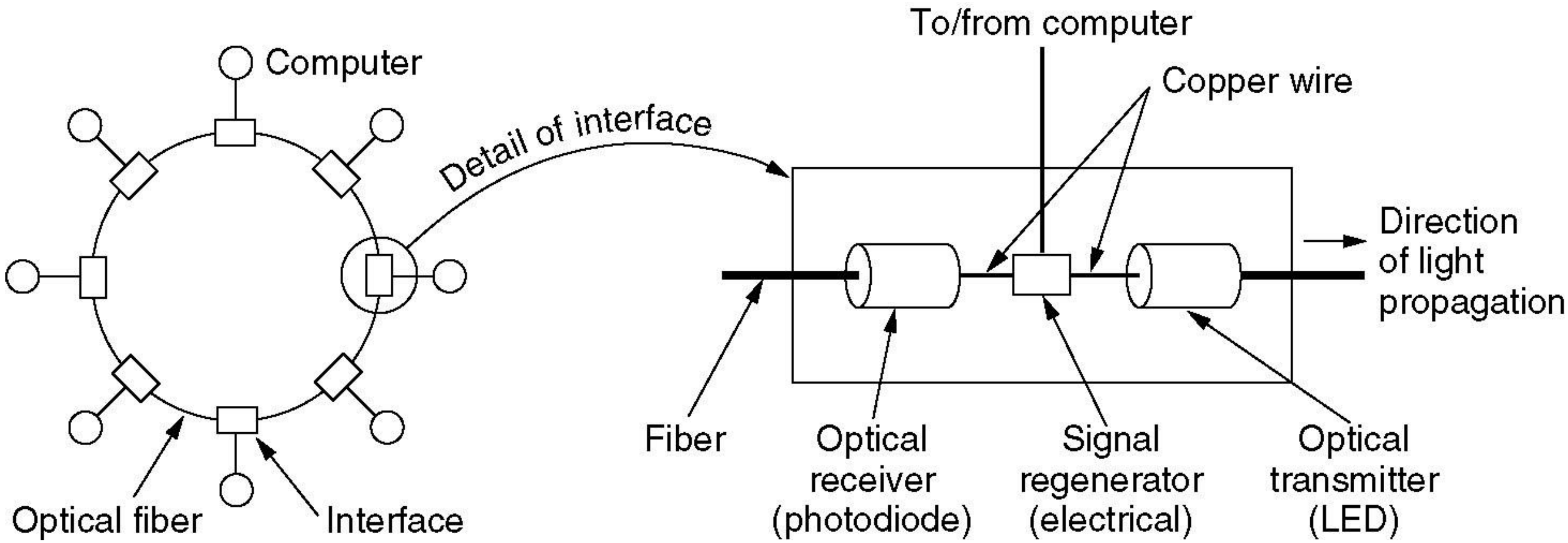


(a) (b)

Attenuation of light

- Attenuation of light thru glass depends upon the wavelength of the light and the physical properties of the glass.

Fiber Optic Networks



Fiber Optic Networks (2)

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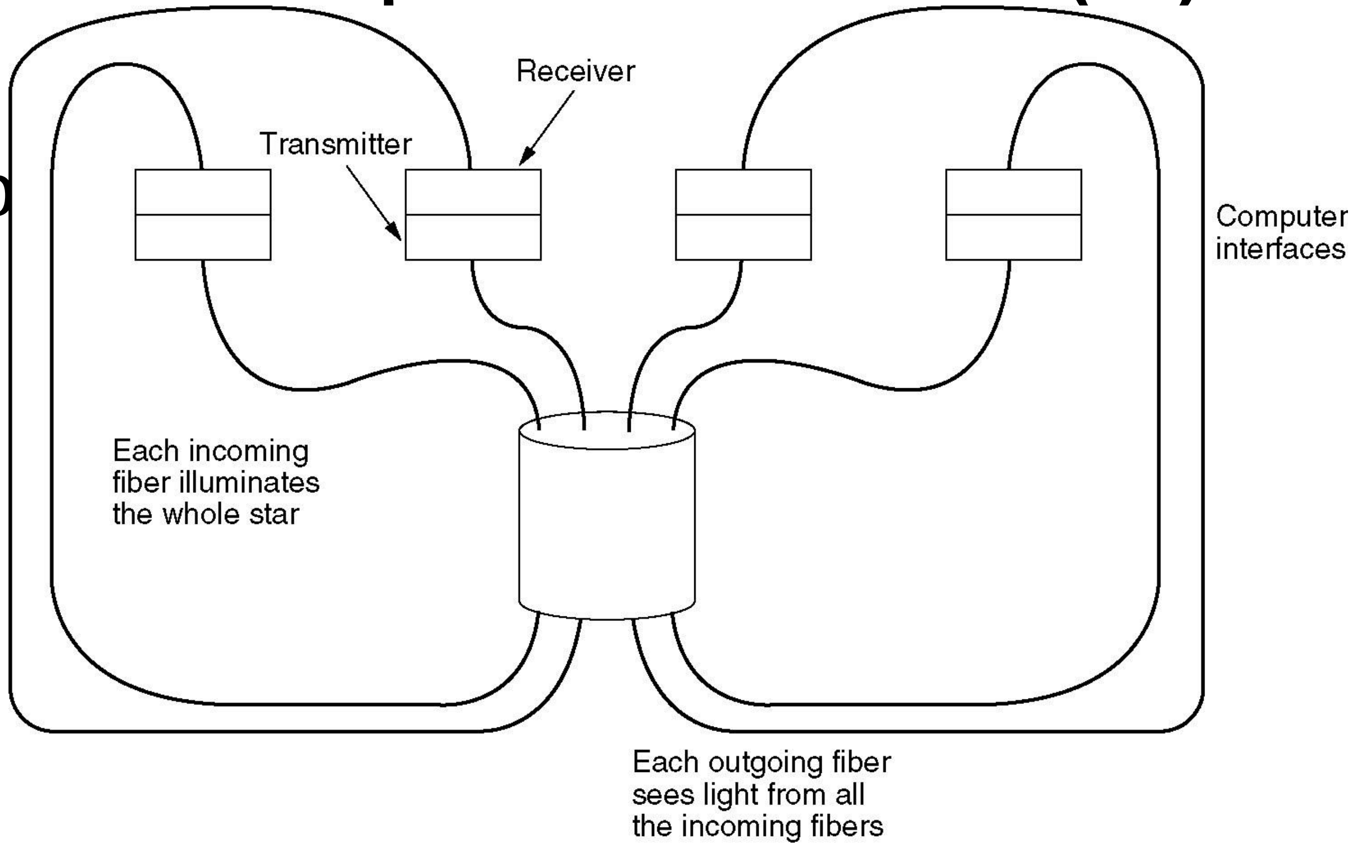

Copper vs Optical

- Adv of fiber
  - High bandwidth
  - Low attenuation, hence repeaters reqd at about 50km vs about 5km for copper – cost saving
  - More immune to external disturbances
  - Lighter than copper
  - Do not leak light and are difficult to tap – security
- Disadv : interfaces and hence maintenance are expensive

Wireless Transmission

- When electrons move, they create electromagnetic waves that travel thru space.

- When an antenna of appropriate size is attached to an electrical circuit, the electromagnetic waves can be broadcast efficiently and received by a receiver some distance away. All wireless communication is based on this principle.

Wireless Transmission

- The Electromagnetic Spectrum
- Radio Transmission
- Microwave Transmission
- Infrared and Millimeter Waves
- Light-wave Transmission

can all be used for transmitting information

Higher Frequency waves

- UV, X-ray and gamma rays can carry more information but,
  - They are hard to produce and modulate
  - Do not propagate well thru buildings
  - And, are dangerous to living things

The Electromagnetic Spectrum

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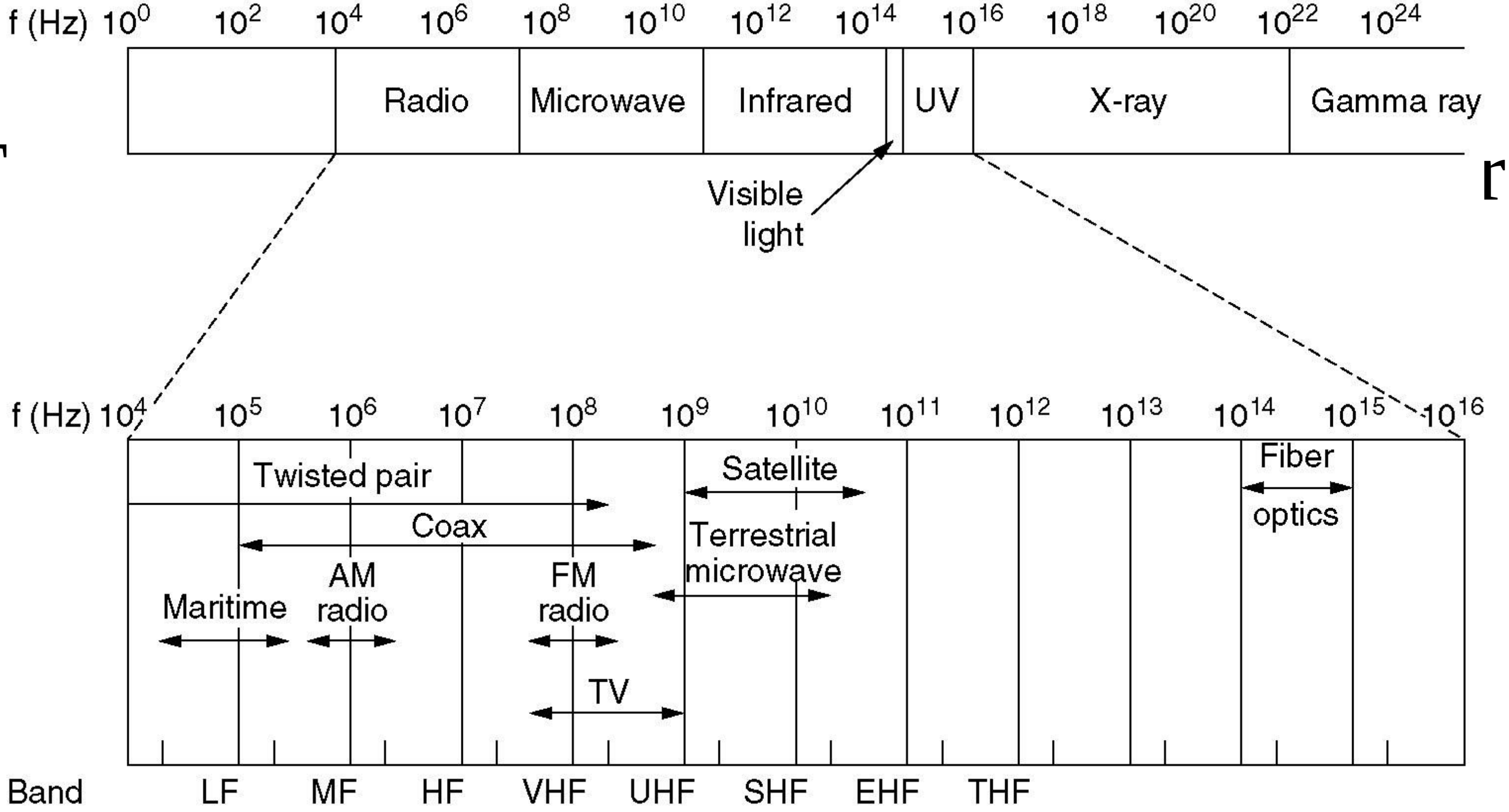

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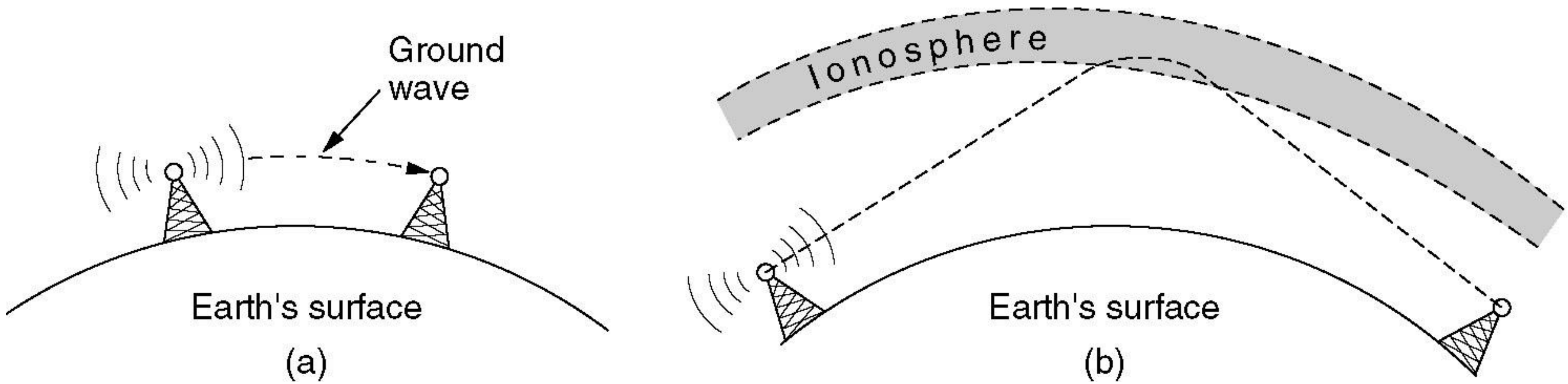
Radio Waves

- Are easy to generate , can travel long distances and can penetrate buildings easily.
- Are omnidirectional,I.e. they travel in all directions
  - Adv : transmitter and receiver do not have to be aligned
  - Disadv : interference of signals : less secure : govt license required to use particular frequency band

Radio waves contd…

- The properties of RW are frequency dependent
  - At low frequency : pass thru obstacles well but the power falls off sharply with distance from the source.
  - At high frequency : they travel in straight lines , bounce off obstcales, and absorbed by rain

Radio Transmission



(a) In the VLF, LF, and MF bands, radio waves follow the curvature of the earth.

(b) In the HF band, they bounce off the ionosphere.

Microwave transmission

- Above 100 MHz, the waves travel in nearly straight lines .
- They do not pass thru buildings well
- Concentrating all energy into a single beam gives a much higher SNR (signal-to-noise ratio) but,
- The transmitting and receiving antennas must be aligned properly.

Microwave transmission

- Since MW travel in a straight line, if the towers are too far apart, the earth will get in the way, hence
- Repeaters are required periodically.
- The higher the towers are, the farther apart they can be.

Multi-path fading in MW

- Even though MW travel in a straight line .. there is some divergence in the space.
- Some waves may be refracted off low-lying atmospheric layers and may take slightly longer to arrive than the direct waves. The delayed waves may arrive out of phase with the direct wave and thus cancel the signal. This effect is called multi-path fading.
- It is a serious problem and is weather and frequency dependent.
- The only solution is to do away with such frequencies and keep some frequencies spare to be used under such circumstances.

MW vs Fiber

- No right of way is needed for MW.
- MW is relatively inexpensive as compared to fiber.

Application of MW

- Short range Networking
- Example : WLL : Wireless Local Loop

Infrared and Millimeter waves

- For short range
- Directional
- Do not pass thru solid objects
- Because of above properties .. No eavesdropping .. Hence secure .. No govt license reqd

Application of Infrared

- Applications of Infrared
  - Remote control Home- appliances
- Applications of Millimeter
  - Wireless Local Loop

Completely different approach

- A completely different approach to allocating frequencies is not to allocate at all. Let everyone transmit at will but regulate the power so that stations have such a short range that they do not interfere with each other.

The ISM(Industrial, Scientific and Medical) band

- Low power, hence short range so that no interference from each other
- For unlicensed usage :
  - Garage door openers,
  - Cordless phones,
  - Radio-controlled toys,
  - Wireless mouse,
  - And numerous other wireless household devices use the ISM band

I Acknowledge

Help from the following site

http://www.cs.vu.nl/~ast/

In preparing this lecture.